Kartik Loya

Ph.D. Candidate in Mechanical Engineering

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Summary

Ph.D. candidate in Mechanical Engineering focused on physics-informed, data-driven modeling and control methods for unconventional robotic platforms, including off-road ground vehicles on deformable terrain and fish-like robots with internal actuation. I genuinely enjoy building and testing robots, and I am especially interested in autonomy that works outside the lab in messy, real environments. My research blends first principles dynamics, multi-modal sensing, and data-driven system identification to build control-oriented predictive models, and I use optimization and learning based control to translate these models into reliable, real-time robot behavior under uncertainty.

Education

Clemson University, PhD in Mechanical Engineering – Clemson, SC, USA Jan 2021 – Jan 2026

- GPA: 4.00/4.00
- Advisor: Dr. Phanindra Tallapragada
- Dissertation: Physics-Informed Modeling, Learning and Control of Robots in Complex Physical Environments

Clemson University, MS in Mechanical Engineering – Clemson, SC, USA Jan 2018 – Jan 2021

- GPA: 3.87/4.00
- Advisor: Dr. Mohammad Naghnaeian

Maharashtra Institute of Technology, Pune University, BE in Mechanical Engineering – Pune, India Jan 2014 – Jan 2018

Research Experience

Graduate Research Assistant, Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS) – Clemson, SC, USA Sept 2024 – present

- Developed a physics-informed, data-driven Koopman learning framework for off-road vehicle mobility on deformable terrain, enabling control-oriented prediction without expensive online terramechanics evaluations.
- Identified soil-specific lifted linear predictors using scalable recursive subspace identification, with data prioritization to capture regime changes from large simulation datasets.
- Integrated the learned predictor into constrained MPC to achieve reliable real-time tracking of aggressive trajectories under steering and torque limits.

Graduate Research Assistant, Applied Dynamical Systems Lab – Clemson, SC, USA Aug 2021 – May 2024

- Developed a linear predictor for high-dimensional nonlinear systems using Koopman operator and subspace identification methods.
- Demonstrated effective control in systems like the hydrodynamic Chaplygin sleigh by applying adaptive control for accurate system stabilization and trajectory tracking.
- Created an online learning framework for updating Koopman operators from streaming data across different dynamical regimes using Gaussian processes as basis functions.
- Investigated the roll dynamics of a hydrodynamic Chaplygin sleigh, analyzing how parametric oscillations and nonholonomic constraints affect the stability of bioinspired underwater robots.
- Designed and developed a fish-inspired robot with internal rotor actuation, applying data-driven methods (Koopman operator and RL) for its modeling and control.

Graduate Research Assistant, Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS) – Clemson, SC, USA — Jan 2020 – Aug 2021

- Designed energy-efficient planning and control strategies for UAV fleets in resource-constrained environments.
- Used MPC to enhance UAV energy efficiency, factoring in disturbances, wind effects, and morphed UAV for improved performance.

Work Experience

Engineering Intern, Thermax Ltd. – Pune, India — Oct 2017 – May 2018

- Surveyed and optimized the steel utilization in the plant.
- Developed a Python program for remote online performance analysis for Vapour Absorption chiller plants.
- Designed and manufactured a boiler panel cutting machine to replace existing flame cutting methods, avoiding the development of heat-affected zones.

Engineering Intern, Elite Techno Group – Pune, India — June 2016 – June 2016

- Designed an ATV on SolidWorks and performed crash analysis.
- Manufactured the ATV and tested its performance on a road track.

Publications

Learning Koopman operators for path tracking by a fish-like robot — Jan 2026
Kartik Loya, P. Chivkula, P. Tallapragada
(Submitted to IEEE Transactions on Robotics)

Differentiable Reinforcement Learning for Path Tracking by an Agile Fish-Like Robot — Jan 2026
Kartik Loya, P. Chivkula, R. Varikuti, P. Tallapragada
(IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS))

Koopman Operator Framework for Modeling and Control of Off-Road Vehicle on Deformable Terrain — Jan 2025
Kartik Loya, P. Tallapragada
(ASME Journal of Autonomous Vehicles and Systems (JAVS))

Agile 3-D Motion Primitives in an Underactuated Fish-like Robot — Jan 2025
P. Chivkula, *Kartik Loya*, P. Tallapragada
(IEEE Robotics and Automation Letters)

Parametric roll oscillations of a hydrodynamic Chaplygin sleigh — Jan 2023
Kartik Loya, P. Tallapragada
(Nonlinear Dynamics)

Online learning of Koopman operator using streaming data from different dynamical regimes — Jan 2024
Kartik Loya, P. Tallapragada
(IFAC-PapersOnLine)

Koopman operator-based predictive control uses a data archive of observables — Jan 2023
Kartik Loya, J. Buzhardt, P. Tallapragada
(ASME Letters in Dynamic Systems and Control)

Stabilization of an Inverted Pendulum on a Nonholonomic System — Jan 2022
Kartik Loya, P. Tallapragada
(IFAC-PapersOnLine)

Competitions

AI for Industry Challenge — Intrinsic x Open Robotics – Team ASIMOV May 2026 – Aug 2026

- Ranked 21st globally and advanced to Phase 1 (Top 30 of 160+ teams) in a competition focused on autonomous manipulation of deformable cables and precision connector insertion.
- Developed the manipulation solution using Intrinsic Flowstate, integrating Vision-Language-Action (VLA) models, reinforcement learning, and OpenCV-based perception.
- Designed and tested reusable robot skills for perception and manipulation within the Flowstate workflow.

Honors and Awards

ME Excellence Award - Ph.D. – Clemson University Apr 2026

- Recognized by the Department of Mechanical Engineering for excellence as a Ph.D. graduate student.

Graduate Travel Grant – Clemson Graduate Student Government Jan 2024

- Awarded competitive graduate travel funding to attend IEEE ICRA 2024.

Talks and Presentations

- **SIAM Conference on Applications of Dynamical Systems (DS25)** — Model-based learning of Koopman operator for complex mechanical systems (May 2025)
- **ISTVS 2025 (55th Conference)** — Data-driven Koopman control-oriented learning framework for improved autonomous off-road mobility (Oct 2025)
- **ICRA 2025 Poster** — Batch learning/updates of Koopman operators from streaming off-road vehicle data with terrain dynamics (Summer 2025)
- **ICRA 2025 Late Breaking Results** — Learning PID Gains for Planar Tracking of Bio-Inspired Swimming Robots Using Simplified Models (2025)
- **MECC 2024** — Online learning of Koopman operator using streaming data from different dynamical regimes (2024)
- **MECC 2023** — Koopman Operator Based Predictive Control With a Data Archive of Observables (2023)
- **MECC 2022** — Stabilization of an Inverted Pendulum on a Nonholonomic System (2022)
- **Clemson VIPR Annual Meeting 2025** — Koopman-Based Modelling and Control of Off-Road Vehicles on Deformable Terrain (2025)

Skills

Programming Languages: MATLAB, Python, C++, Julia

Software & Tools: Mathematica, SolidWorks, ROS, ROS2, PyTorch, GPOPS-II, MuJoCo

Research Areas: Physical and Embodied AI, Data-driven modeling and control, Reinforcement Learning, Optimal control, Trajectory optimization, Bio-inspired underwater robots, Off-road vehicle autonomy, Nonholonomic/underactuated systems, Physics-informed learning, Koopman/Operator-theoretic modeling,

Coursework

Graduate Coursework: Nonlinear Dynamics & Chaos | Robot Multibody Dynamics | Linear Analysis,Advanced Nonlinear Control | Random Processes and Application | Applied Optimal Control, Networked Dynamical Systems | Advanced Estimation | Control of Cyber-Physical Systems | Autonomy: Science & Systems | Reinforcement Learning | Advanced Machine Learning | Theory of Elasticity